

NORUN NABI

Doctoral Researcher in Computer Science • Cybersecurity & Machine Learning • FinTech Security Engineer
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RESEARCH PROFILE

Doctoral researcher in Computer Science at the University of the Potomac (GPA 4.00) with about a decade of industry experience building secure FinTech and payment systems. Author of **9 peer-reviewed publications (2024–2025)** in cybersecurity, machine learning, and applied AI, with **223 citations, h-index 6, and i10-index 5** (Google Scholar, September 2026, including an M.Sc. thesis); first author of two, including an IEEE conference paper on insider-threat detection. Invited peer reviewer for *Engineering Applications of Artificial Intelligence* (Elsevier) and *Discover Computing* (Springer Nature). CompTIA Security+ and ISC2 CC certified.

RESEARCH INTERESTS

- Cybersecurity — anomaly detection, insider-threat detection, IoT and satellite network security
- Machine Learning — supervised and unsupervised learning, data analysis, time-series forecasting
- Applied AI — generative AI and large language models (LLMs)
- Quantum Computing

EDUCATION

Doctor of Computer Science (in progress) — University of the Potomac, Washington, DC · Sep 2025 – Present
GPA 4.00 · anticipated completion Dec 2028

M.S. in Information Technology — Washington University of Science and Technology, Alexandria, VA ·
Oct 2022 – Jun 2024

GPA 4.00 / 4.00 · Presidential List · Track: Information Technology Systems and Management

M.Sc. in Computer Science and Engineering — BRAC University, Dhaka, Bangladesh · 2022

Thesis: A document vectorization approach to Resume Ranking System (RRS) — supervisor Dr. Muhammad Iqbal Hossain

B.Sc. (Honours) in Computer Science and Engineering — Jahangirnagar University, Savar, Dhaka, Bangladesh ·
2008 – 2011

CGPA 3.22 / 4.00

PEER-REVIEWED PUBLICATIONS

1. Mahin, M. R. H., Ahmed, E., Mukti, M. H., Rana, M. N. U., Islam, M. T., **Nabi, N.**, Akhi, S. S., & Shaima, M. (2025). Real-time Twitter trending topics detection and sentiment analysis using data mining technique. In *2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)* (pp. 72–76). IEEE. <https://doi.org/10.1109/RAAICON69033.2025.11502118>
2. **Nabi, N.**, Tuhin, M. K., Bashir, M., Lucky, K. Y., Raihan, M., Imam, H., & Ansar, M. T. B. (2025). Continual learning pipelines for detecting insider threats across corporate cybersecurity infrastructures. In *2025 International Conference on Electrical Engineering and Informatics (ICEEI)* (pp. 1–8). IEEE. <https://doi.org/10.1109/ICEEI68459.2025.11330487> [Cited by 8]
3. Sharmin, A., Mahmud, B. U., **Nabi, N.**, Shaima, M., & Faruk, M. J. H. (2025). Cyber attacks on space information networks: Vulnerabilities, threats, and countermeasures for satellite security. *Journal of Cybersecurity and Privacy*, 5(3), 76. MDPI. <https://doi.org/10.3390/jcp5030076> [Cited by 19]
4. Ahmed, E., Shaima, M., Tusher, M. I., **Nabi, N.**, Rana, M. N. U., & Saha, S. (2025). Health care — An Android application implementation and analyzing user experience. In *2025 IEEE 5th International Conference on Smart Information Systems and Technologies (SIST)* (pp. 1–6). IEEE. <https://doi.org/10.1109/SIST61657.2025.11139168>

5. **Nabi, N.**, Rahman, M. M., Ghosh, S. K., Alam, S., Barua, R., Asaduzzaman, M., & Shatu, N. R. (2024). Machine learning-based anomaly detection for cyber threat prevention. *Journal of Primeasia*, 6(1), 1–8. <https://doi.org/10.25163/primeasia.6110172> [Cited by 37]
6. Sayed, M. A., Badruddowza, Sarker, M. S. U., Al Mamun, A., **Nabi, N.**, Mahmud, F., Alam, M. K., Hasan, M. T., Buiya, M. R., & Choudhury, M. Z. M. E. (2024). Comparative analysis of machine learning algorithms for predicting cybersecurity attack success: A performance evaluation. *The American Journal of Engineering and Technology*, 6(9), 81–91. <https://doi.org/10.37547/tajet/Volume06Issue09-10> [Cited by 17]
7. Shaima, M., Rana, M. N. U., Islam, M. T., **Nabi, N.**, Tusher, M. I., Ahmed, E., Saha, S., & Al Shiam, S. A. (2024). Review on blockchain for IoT security and data integrity. *Computer Science & Information Technology (CS & IT)*, 14(11), 115–126. <https://doi.org/10.5121/csit.2024.141110>
8. Linkon, A. A., Shaima, M., Sarker, M. S. U., Badruddowza, **Nabi, N.**, Rana, M. N. U., Ghosh, S. K., Esa, H., & Chowdhury, F. R. (2024). Advancements and applications of generative artificial intelligence and large language models on business management: A comprehensive review. *Journal of Computer Science and Technology Studies*, 6(1), 225–232. <https://doi.org/10.32996/jcsts.2024.6.1.26> [Cited by 100]
9. Shaima, M., **Nabi, N.**, Rana, M. N. U., Islam, M. T., Ahmed, E., Tusher, M. I., Mukti, M. H., & Saad-Ul-Mosaher, Q. (2024). Elon Musk's Neuralink brain chip: A review on 'brain-reading' device. *Journal of Computer Science and Technology Studies*, 6(1), 200–203. <https://doi.org/10.32996/jcsts.2024.6.1.22> [Cited by 41]

THESIS

- **Nabi, N.** (2022). *A document vectorization approach to Resume Ranking System (RRS)* [Master's thesis, BRAC University; supervisor: Dr. Muhammad Iqbal Hossain]. <http://hdl.handle.net/10361/17980> [Cited by 1]

PEER REVIEW SERVICE

- Reviewer, *Engineering Applications of Artificial Intelligence* (Elsevier) — manuscript EAAI-26-3522, 2026
- Reviewer, *Discover Computing* (Springer Nature) — 2026

PROFESSIONAL EXPERIENCE

9th Networks Sep 2024 – May 2025

DevOps Engineer

- Built and maintained CI/CD pipelines, improving deployment efficiency and shortening feedback loops for development teams.
- Designed containerization strategies with Docker, improving resource utilization across application environments.
- Automated infrastructure deployment and monitoring to improve system reliability and scalability.
- Monitored automated build and integration processes; led root-cause investigation and resolution of build/release failures.
- Facilitated team training sessions on modern DevOps practices, raising team productivity and tooling proficiency.
- Partnered with development and QA in an agile environment to drive continuous improvement of cloud-based delivery.

Dev Environment: Python, AWS, Docker, Git, Jenkins, Bamboo, Linux

Pubali Bank PLC — pubalibangla.com May 2020 – Oct 2022

Senior Software Engineer / Technical Lead

- Architected and developed cloud-based API services using Java, Spring Boot, Spring Cloud, Spring Gateway, Spring Security, Hibernate, and RESTful web services, deployed against PostgreSQL and AWS RDS.
- Implemented microservices and CQRS patterns; integrated Keycloak (OAuth 2.0/OIDC) for identity, Kafka for event streaming, and webhooks for system integration.
- Designed and delivered Angular 15 web applications using NgRx, Webpack 5 Module Federation, Angular Material, and Bootstrap — focusing on state management and modular, maintainable architecture.

- Developed identity, notification, and scheduling services with Spring Security, JWT, and Quartz Scheduler in a multi-tenant database design.
- Containerized services with Docker and deployed on Kubernetes; established CI/CD with Jenkins and runtime monitoring with Prometheus.

Dev Environment: Java, Spring Boot, Spring Cloud/Gateway/Security, Hibernate, Keycloak, PostgreSQL, AWS RDS, Kafka, Jenkins, Prometheus, TypeScript, Angular 15, NgRx, Webpack, Node.js, Docker, Kubernetes

Kona Software Lab — konai.com Dec 2014 – Mar 2020

Senior Software Engineer

- Engineered Kona Pay-PaaS, a microservice-driven Payment Platform as a Service (PaaS) for processing financial transactions in compliance with Visa, Mastercard, and Diners Club standards.
- Developed Kona Money, a scalable mobile payment platform leveraging HCE SE (Secure Element) and CardSE; implemented Visa/Mastercard HCE specs enabling NFC POS, in-store, QR, and barcode payments via the mobile wallet.
- Built a key management system (K-KMS) in Java/Spring with MyBatis and MySQL to securely manage cryptographic keys and certificates, hardened with encryption, access controls, and ISO security standards.
- Developed K-CPS, a web-based EMV ICC personalization solution, and K-PAS for automated card personalization profile generation following Global Platform (GP) specifications.
- Directed K-IIS to enable instant card issuance within bank branch offices, integrating with existing card issuance infrastructure.
- Deployed and managed microservices on Wildfly and Tomcat; integrated Spring Cloud for centralized configuration, service discovery, and load balancing.
- Implemented Spring Batch jobs to process large volumes of financial data; automated builds, tests, and deployments with Jenkins and Apache Maven.
- Collaborated with cross-functional teams to gather requirements, architect solutions, and deliver on schedule.

Dev Environment: Java, Spring Boot/Cloud, JPA/Hibernate, MyBatis, Maven, Gradle, RESTful APIs, Microservices, RabbitMQ, Liquibase, Wildfly, MySQL, Oracle, Jenkins, Git, Agile (Scrum)

MB Software Mar 2014 – Dec 2014

Android Developer

- Implemented an Android reservation-system application following platform UI guidelines.
- Integrated social-media SDKs and app analytics.

Dev Environment: Java, Android SDK, Swagger API

BJIT — bjitgroup.com Mar 2013 – Mar 2014

Programmer (Java)

- Designed and implemented RESTful Web APIs to streamline client-server communication and enable reliable data flow.
- Managed critical financial data on the backend database with attention to data integrity and scalability.
- Enhanced user interfaces with interactive features using JavaScript and jQuery.

Dev Environment: Java, Java Swing, PostgreSQL, RESTful APIs, JavaScript, jQuery

CERTIFICATIONS

- CompTIA Security+ (SY0-701) — CompTIA, 2025 (valid through Oct 2028)
- ISC2 Certified in Cybersecurity (CC) — ISC2, 2025 (valid through Jul 2028)
- Google Cybersecurity Professional Certificate — Google, 2025
- DevOps Engineering with Cloud Computing — PeopleNTech, 2024
- ITEE Fundamental Information Technology Engineer (FE) — ITPEC, 2022 — [verification](#)
- Natural Language Processing with Classification and Vector Spaces — DeepLearning.AI (Coursera), 2021

AWARDS & HONORS

- Presidential List (Presidential Excellency Award) — Washington University of Science and Technology, 2024, for a 4.00/4.00 GPA
- Managing Director’s Recognition — Kona Software Lab, 2019, for contribution to “Nagad,” a mobile financial service (MFS) platform used by millions in Bangladesh

PROFESSIONAL MEMBERSHIPS

- IEEE Computer Society — Member
- ISC2 — Member (Certified in Cybersecurity), 2025 – 2027

TECHNICAL SKILLS

Security	PKI, certificate authority, key management, symmetric/asymmetric encryption, hashing, tokenization, MFA, zero trust, IDS/IPS, WAF, SIEM, Wazuh, Nmap, Wireshark, OSINT, OAuth 2.0, OIDC, JWT, Keycloak, TLS
ML & Data	Scikit-learn, PyTorch, TensorFlow, Keras, Pandas, NumPy, SciPy, OpenCV
Languages	Java, Python, TypeScript, JavaScript, Bash
Backend	Spring Boot, Spring MVC, Spring Cloud, Spring Gateway, Spring Security, Spring Data (JPA, Redis, Mongo, Elasticsearch), Spring Batch, Hibernate, Project Reactor, Node.js, Nest.js, OAuth 2.0, JWT, Active Directory, Kafka Streams, ActiveMQ, Resilience4j, Hystrix, CQRS, SAGA, Maven, Gradle
Cloud & DevOps	AWS (IAM, EC2, RDS, S3, SNS, SES, VPC, SQS, Route 53, EKS, ECS, ECR, Lambda, Fargate, CloudFront, CloudWatch, CloudFormation, Elastic Beanstalk), Docker, Docker Compose, Docker Swarm, Kubernetes, KEDA, Terraform, Nginx
CI/CD	GitHub Actions, Jenkins, Bitbucket Pipelines, Atlassian Bamboo, AWS CLI, Shell scripting
Databases & Cache	PostgreSQL, MySQL, Oracle DB, MongoDB, Neo4j, H2, Elasticsearch, Redis, Memcached, Varnish, Flyway, Liquibase
Observability	Prometheus, Grafana, Node-Exporter, Zabbix, ELK Stack
Architecture	Microservices, Event-driven, Clean Architecture, SOLID, OOP, GOF patterns, Dependency Injection, IoC, TDD, BDD

PROJECTS

Notable NLP / ML Project — RRS (Resume Ranking System)

- Built a domain-trained word2vec / TF-IDF model specialized on resume vocabulary; used cosine similarity to match resumes to job descriptions and k-nearest neighbors to cluster candidate documents.

Dev Environment: Python, FastAPI, React

Notable Docker Project — Workspace Orchestration

- Designed a self-hosted, offline-first Docker platform running 13 containers — PostgreSQL 16, Plane (project management with MinIO object storage), Odoo 19, Redis, RabbitMQ, Portainer, and an Alloy / Loki / Grafana logging stack — on a shared Docker network with static IPs.
- Built one-command startup and shutdown orchestration in Bash: health-gated service ordering, graceful PostgreSQL stop, pg_dumpall and MinIO volume backups with gzip integrity checks, retention pruning, and an off-site Google Drive mirror via rclone that refuses to sync on corrupt or empty local state.
- Developed FastAPI-based management agents with web dashboards, and a mirrored logging convention covering 125+ scripts so every run is traceable in Grafana Loki.

Dev Environment: Docker, Docker Compose, Bash, Python, FastAPI, PostgreSQL, Nginx, Grafana, Loki, Alloy, rclone